

Ashwin Conrad

Mechanical engineering student with hands-on experience in vehicle low-voltage systems, industrial panel assembly, engineering analysis, and technical automation. Interested in automotive, motorsport, manufacturing, test, and product-development engineering roles.

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EDUCATION

University of Alberta	Mechanical Engineering - Edmonton, AB	2023 - Present
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WORK EXPERIENCE

AltaGas - Calgary, AB Integrity & Reliability Engineering Student	2025 - 2026
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- Developed Python, Microsoft Fabric, and Power BI workflows for corrosion monitoring, inspection planning, and asset lifecycle work across multi-facility operations.
- Analyzed 2+ years of operating data for 12+ brazed aluminum heat exchangers, screening against OEM limits and summarizing monitoring gaps and risk-based recommendations.

Spartan Controls - Calgary, AB Control Panel Technician	Summer Terms, '23, '24
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- Built, wired, loaded, and checked 30+ industrial control panels using engineering drawings, electrical schematics, pre-delivery QA, and documented deficiency resolution.
- Processed 250+ PLC software updates in under half the allocated time while supporting panel production and quality checks.

LEADERSHIP	COMMUNITY INVOLVEMENT
Formula EV Racing - Low Voltage Electronics Lead Lead low-voltage electrical-system planning and integration for Formula UA27, coordinating board-level hardware, safety interfaces, sensors, power distribution, and vehicle packaging across mechanical and software teams.	CYDC Basketball Coach Youth coaching and team development.
Engineering Student Engagement Leader 2024-2025	

GENERAL SKILLS		
Mechanical Design	Project Management	Electrical Panel Design
Low Voltage Electronics	Data Analysis	Fabrication & CAD

RECOGNITION		
Undergraduate Student Leadership Award, 2024-2025	Leadership in Engineering Scholarship, 2023-2024	APEGA Science Olympics Medalist: Silver, 2021-2023

SELECTED PROJECT PORTFOLIO		
Heat Exchanger Lifecycle Analysis AltaGas		2025 - 2026
<i>Python, data analysis, technical assessment</i>		
Analyzed 2+ years of operating data for 12+ brazed aluminum heat exchangers to assess fatigue risk and monitoring gaps. Summarized lifecycle findings into practical recommendations for inspection planning and operating awareness.		
Inspection Scope Generation Tool AltaGas		2025 - 2026
<i>Python, openpyxl, python-docx</i>		
Developed a Python tool to generate 500+ equipment-specific inspection scopes from database exports. Reduced manual document creation time by standardizing Excel-to-Word scope generation.		
Formula EV Electrical Systems Formula EV Racing		Winter 2025 - Present
<i>Low-voltage electrical systems</i>		
Coordinate low-voltage power distribution, sensor integration, and safety-interface planning around the UA27 vehicle package. Work across mechanical and software constraints to make electrical systems accessible, serviceable, and competition-ready.		
Leather Tool Carrier Spartan Controls		2024
<i>Shop workflow tooling</i>		
Designed and built a leather tool carrier to keep common hand tools organized and accessible during panel assembly. Turned a personal workflow problem into a durable shop aid for faster repeated work.		

TECHNICAL SKILLS	
CAD & Design	SolidWorks, AutoCAD & 2D laser cutting, engineering drawings, mechanical prototyping, 3D modelling & printing
Vehicle & Electrical	low-voltage systems, power distribution, sensor integration, electrical schematics, panel assembly, QA
Data & Automation	Python, Excel data analysis, Power BI, Microsoft Fabric, Power Query, openpyxl, python-docx
Engineering Software	Aspen HYSYS, Maximo, JDE
Fabrication	Metal, leather, composite, and small mechanical build projects